

Philosophy 415: Final Exam

Due on Friday, December 13, 2019

University of New Mexico 5:00 PM

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1

Frege's initial axiomatization of set theory included a "Universal Comprehension Principle" as an implication of his "Basic Law V". Please explain what this principle was and why and how it was modified in the eventual ZFC axiomatization of set theory. Do the axioms of ZFC solve the underlying problem with Frege's principle, in your opinion? Why or why not?

The "Universal Comprehension Principle" is Frege's "Basic Law V" Principle. In Frege's time, he referred to it as the Basic Law V Principle; however, it evolved into the Universal Comprehension Principle which is what we call it today. This principle states that for any property definable in language, there is a set consisting of all and only elements that have that property. At the time, this included sets that are members of themselves (also known as reflexive or self-membered sets). For instance, if we consider:

and we ask: is P discussed on this page? Then we can say yes because we are talking about P now. Therefore, since we are talking about P , which is an element of this page, we are saying $P \in P$. However, this changed once Russell proposed his famous paradox regarding the set of all sets that are not elements of themselves. This is written as (where R stands for Russell's set):

$$R = \{x \mid x \notin x\}$$

In other words, R is not reflexive. Now we have to ask, is R an element of itself or not? If it is an element of itself, then it is reflexive. However, the definition of R states that R is not reflexive. Therefore, we have $R \notin R$ which contradicts our original assumption that $R \in R$. Now, let's assume the opposite that $R \notin R$. If that is the case, then R is not reflexive. However, by R 's definition, this makes $R \in R$. This again leads us to a contradiction from our first assumption that $R \notin R$. Either option, leads to paradox of oscillating truth values which Russell communicated to Frege. This would later go on to be known as the Vicious Circle Principle.

Why does this matter? It matters because the Basic Law V states that knowledge of a natural link between language and existence can exist in an actual object of thought. Russell's paradox seems to show that this isn't true, and that the Russell set must necessarily not exist. When it does, it leads to paradoxical implications pertaining to language involving reference to the totality and the self.

ZFC's axiomatization of set theory attempted to mitigate this by implementing the Schema of Specification. The Schema of Specification states that given any set A and any well-defined property ψ , the set B of all elements of A having the property ψ exists. This in conjunction with the Axiom of Foundation is a two-fold way to prevent Russell's paradox and oscillating truth values because it says that we can't create a set of sets.

Moreover, it also paints a picture of the set theoretical universe called the iterative conception of sets also known as Big V. In my opinion this "Big V" solves the problems presented by Russell's paradox because by accepting the underlying intuitions of "Big V" we prevent self-membered sets and the set of all sets. In other words, we avoid the problem with self-reflexivity and totality.

2

Please state and explain briefly the Continuum Hypothesis in its general form (GCH). We now know that the GCH cannot either be proven or refuted from the ZFC axioms. How do we know this?

The Continuum Hypothesis in its general form is:

For every ordinal α , $2^{\aleph_\alpha} = \aleph_{\alpha+1}$

$2^{\aleph_\alpha}$ is the next cardinality of real numbers after aleph-null which cannot be put into a one-to-one correspondence with the set of all-natural numbers. The Continuum Hypothesis states that the cardinality of the power set for Ω (cardinality of real numbers) is equal to $\aleph_2$. In other words, we don't know the extent of the "power" set operation and how much it exceeds the set itself.

The Continuum Hypothesis cannot either be proven or refuted from the ZFC axioms because it is independent of ZFC. We know this from Gödel's second incompleteness theorem which proves the GCH cannot be disproven from the axioms. If we assume that ZFC is consistent, then we know ZFC and the GCH are consistent. Gödel proved this, along with axiom of choice, using his "inner models" method which showed that in universe L (Constructible universe), GCH holds true. This means that within model M, L is a sub-structure universe of ZF which GCH holds true. In other words, that it is impossible to prove the negation of GCH.

This in combination with Cohen's "forcing" proved that the negation of GCH added to ZFC does not lead to a contradiction. For instance, if we assume that ZFC is consistent, then ZFC and not GCH is consistent. He does so using his "outer model" which shows that in a separate constructible model that has more sets than the original model GCH does not hold true. This means that GCH is equally possible to be false in ZFC. Therefore, also proving that GCH cannot be proven from ZFC.

3

With reference to Russell's paradox and theory of types, explain what Priest calls "parameterization" and why he thinks this is an inadequate solution, both to Russell's paradox and to a series of related problems. Do you agree or disagree? How does Priest's skepticism about parameterization help to support his dialetheism?

Priest's procedure of indexing totalities such that we deny their existence and refrain from referring to them so we can avoid contradictions is known as "parameterization". This is a solution to the problem both Russell and Zermelo-Fraenkel faced when grouping together all the sets or set of all sets which do not contain themselves. Russell tried to rectify this paradoxical problem with his theory of types (ZF did so using the axioms of foundation and separation). Russell classified sets into levels which could only be formed at a definite level, preventing any set from including itself as an element. This implies that the formation of classifying sets will go on indefinitely with no end thereby preventing the formation of the set of all sets not containing themselves.

Moreover, Priest believes that "parameterization" is an adequate solution to paradoxes of this kind because it lets us see what is truly happening in a simple way. For instance, when looking at "set-theoretical" paradoxes (involving self-membered totalities), parameterization lets us see the general form and structure the arguments take. Thereby giving us insight to the same thought problems, which occurs when discussing self-membered totalities, philosophers encounter frequently.

The implication of parameterization is that for our set-theoretical systems, which include our systems for thinking about the limitation of thought and language, contain inconsistencies. These inconsistencies imply that we have statements within our thought and language of: $P \ \& \ \sim P$. Priest's dialetheism accepts these inconsistencies; however, this doesn't imply anything can be derived from the system. It means that we can accept some inconsistencies by modifying the conditional into a disjunction. Priest call this dialetheism.

Priest's skepticism about parameterization helps support his dialetheism because he is skeptical about the system itself and whether it can build a system which tolerates contradictions. If so, then we can prevent contradictions from multiplying, since we can characterize them using the inclosure schema, and continue to reason about set-theoretical problems.

Personally, I think having any inconsistencies damages the system. By allowing and even building a system which allows this is a tough no for me.

4

Please explain how Gödel's second incompleteness theorem follows from his first. Given Gödel's second incompleteness theorem, do you think we should be skeptical about the consistency of any or all of our various formal systems for arithmetic (ZFC, PM, Peano Arithmetic, etc.?) Why or why not?

Gödel's second incompleteness theorem follows from his first because his first incompleteness theorem proves that for any sufficiently strong system which is Ω consistent, there exists a sentence that is syntactically undecidable such that the system is unable to prove it or the negation of it. This ultimately disproves Hilbert's formalist plan. However, we can still try to solve his second goal, which is to have a formal system that proves its own consistency. This is essential because if not, one formal system will need to rely on another system which means that that system also needs to rely on another system and so on to prove any formal systems consistency.

Thankfully, Gödel's first incompleteness theorem provides enough resources that we can prove the second easily. This is so because from Gödel's first incompleteness theorem we have:

$$\vdash PM \text{ Con}(PM) \implies \neg Prov(\#GS)$$

Which means we also have:

$$\vdash PM \text{ Con}(PM) \implies GS$$

And if that were true that we had a proof for the consistency of PM or any other system like it, then we would also have a proof of GS . However, we know that we don't because of Gödel's first incompleteness theorem. Therefore, we don't have a proof for the consistency of PM thus giving us Gödel's second incompleteness theorem. That is, for any system (let's say R fl), with the ability of modeling arithmetic, if R is consistent then R cannot prove its own consistency.

Gödel's second inconsistency seems to imply that we should be skeptical of any formal system modeling basic arithmetic since it can't prove its own consistency. In a way this is true and in a way it is not. Ideally, we want a system who can prove the consistency of itself, but because we cannot find one it doesn't necessarily mean that we can't find another way to prove the consistency of our system. It just means that we can't prove its consistency in that way.

5

The thesis of “mechanism” holds that the human mind is essentially a Turing machine. With reference to Lucas’s argument, do you think that Gödel’s incompleteness theorems provide a convincing case against mechanism? Why or why not?

Gödel’s incompleteness theorems do not provide a convincing case against mechanism, in reference to Lucas because they don’t apply. Lucas argues that Gödel’s incompleteness theorems prove anti-mechanism, but we know that isn’t true because those theorems only apply to consistent systems and humans are not consistent systems. Therefore, it doesn’t apply here.

Lucas can argue that our inconsistencies equate to the same inconsistencies of the occasional malfunctioning of a machine and not the normal set rules, but how does he know that? If he wanted to say this, then he would need to provide a consistent system of logic which equates to that of human’s system of logic to prove this.

6

[Extra Credit 1]. In a footnote to the famous 1931 paper in which he announced his two ‘incompleteness’ theorems, Gödel wrote: “The true reason for the incompleteness which attaches to all formal systems of mathematics lies ... in the fact that the formation of higher and higher types can be continued into the transfinite... while, in every formal system, only countably many are available. Namely, one can show that the undecidable sentences which have been constructed here always become decidable through adjunction of sufficiently high types.” Why do formal systems only make available “countably many” types (in the sense of Russell’s theory of types)? How can the “formation” of higher transfinite types take place, anyway? What is Gödel suggesting, then, about how his incompleteness theorems are related to Cantor’s transfinite hierarchy?

Formal systems only have finitely many types because everything in the system is defined by a string of symbols in the system. Gödel showed that there are countably many of these statements that can be made. Therefore, there are countably many statements in the system, which means that there are countably many types.

If you had a system with a countably infinite number of types, then you could declare a type for each element in the power set of the set of this languages types. Then, we could base a new language off this set of types and iteratively continue this process to create higher order transfinite types.

Gödel is suggesting that his incompleteness theorems are the same way that for any infinite set, elements present in a later set can be demonstrated or generated with diagonalization in the same way that for any formal system a statement independent of the system may be made using diagonalization.

7

[Extra Credit 2]. In what ways are the formal and structural issues underlying Russell's paradox and Gödel's theorems similar? In what ways are they different? Are there any general lessons we can draw from both results about thought and the objects of thought? What are these lessons, if so?

Russell's paradox arises from a self-reference issue whereas Gödel's arises when you say something like P cannot be proven. Russell's paradox stems from questions regarding membership whereas Gödel's stems from questions regarding provability and truth. They are similar in that they both can be employed using a diagonal argument to make their claims. Gödel also deals with self-reflexivity issues; however, he uses Gödel numbering to mitigate that and ask whether the sentence can be proven from the axioms.

As a general lesson, we can learn from both of these results is that when we are reasoning about ourselves our reasoning will either have inconsistencies or never fully be complete.